

COVID-19 Global Analysis

This project explores **how COVID-19 spread across the world using data from Our World in Data**. I focused on three simple but interesting questions:

1. When did each country report its **first confirmed case**?
2. When did each country hit its **peak in daily new cases**?
3. How did the **Case Fatality Ratio** (CFR) change over time?

The analysis is done in Python using Pandas and Matplotlib, inside a Jupyter Notebook.

Dataset

Source: **Our World in Data / COVID-19 dataset**

Format: CSV

Scope: Daily records for each country, including cases, deaths, testing, vaccination, and demographic information.

After exploring the dataset, the following columns were selected for the core analysis:

- continent
- country
- date
- total_cases
- new_cases
- total_deaths
- new_deaths
- population

Data Cleaning

Like most real-world healthcare data, this dataset came with:

- missing values
- inconsistent country names
- entries with zero cases but included in totals

The notebook emphasizes that **data cleaning is not a one-time step** but an ongoing process throughout the analysis.

Key Analyses

First Confirmed Case Timeline

For each country:

- The first date with total_cases > 0 was extracted.
- A timeline scatter plot shows how the virus spread globally.
- Most countries reported cases within the first half of 2020, while many in **Oceania did not report cases until 2021** due to geographical isolation.

Peak Wave Detection

For each country:

- The date of maximum daily new cases (new_cases) was calculated.
- A bar chart shows how many countries peaked per month.
- After cleaning, the majority of peaks clustered around **Dec 2021 – Mar 2022**.

Case Fatality Ratio (CFR) Trends

CFR was calculated as: $CFR = \text{total_deaths} / \text{total_cases}$

Example comparisons were made between:

- Greece
- Italy
- United States

A clear pattern emerged:

- **Italy experienced a much higher CFR in early 2020**, especially during the first wave.
- CFR declined significantly from 2021 onward as pandemic responses improved worldwide.

Visualizations

The notebook includes:

- First-case timeline scatter plot
- Bar chart of monthly peak waves
- CFR trend comparison line plot

All visualizations are generated in Python.

Conclusions

The analysis highlights:

- The rapid spread of COVID-19 in early 2020
- How isolated island nations (especially in Oceania) remained COVID-free the longest
- How peak outbreaks aligned globally around late 2021 – early 2022
- CFR trends demonstrate the dramatic improvement of medical response and detection as the pandemic progressed

The project also reinforces the importance of:

- Constant data validation
- Cleaning throughout the workflow
- Not blindly trusting raw datasets

Tools Used

- Python
- Pandas
- Matplotlib
- Jupyter Notebook
- Our World in Data (COVID-19 dataset)

Contents

- Jupyter Notebook with full code and plots
- Exported PDF with results and commentary
- This README

Next Steps

- Try a SQL version of the same analysis
- Add forecasting models
- Explore vaccination or mobility trends

Contributions

Suggestions, improvements, or pull requests are welcome!